Customer Shopping Behavior Analysis

Insights from 3,900 Transactions: Uncovering Spending Patterns, Customer Segments, and Strategic Opportunities.

Dataset Overview

3.9K
Total Transactions

$59.76
Avg Purchase Amount

3.75
Avg Review Rating

Dataset covers 18 key features including demographics, purchase details (category, size, color), and behavioral data (discounts, frequency, shipping type).

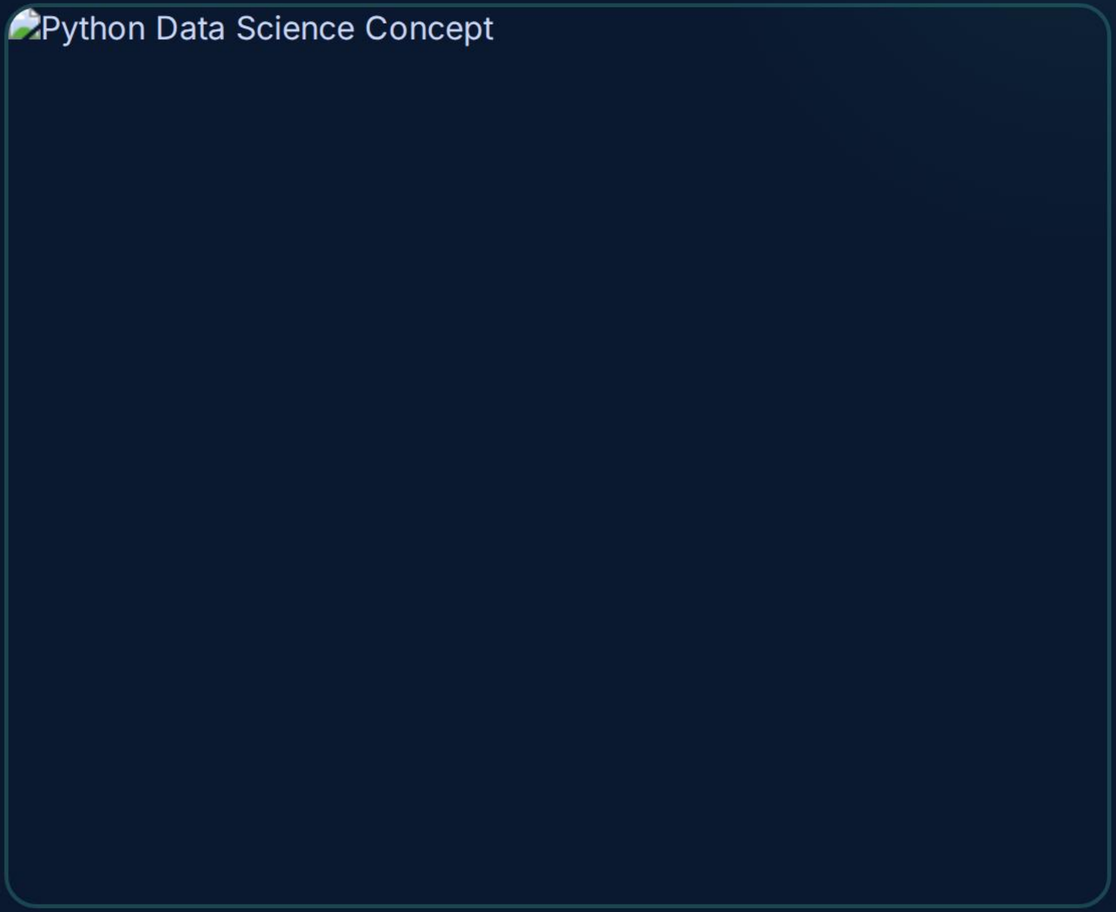
Cleaning & Imputation

Handled missing data in Review Ratings using median category imputation. Standardized columns to snake_case.

Feature Engineering

Created age_group bins and purchase_frequency metrics. Optimized dataset by dropping redundant promo code features.

SQL Integration



Revenue by Gender Analysis

- **Male:** $157,890 (67.7%)
- **Female:** $75,191 (32.3%)

Male customers contribute more than double the revenue of female customers, highlighting a significant demographic segment for targeting.

High-Value Discount Users

Analysis Segment

We identified **839 unique customers** who utilize discounts while maintaining purchase amounts significantly above the corporate average.

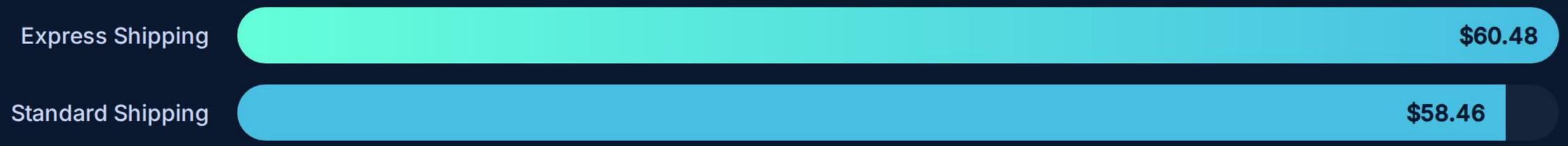
- High Spending Core: Purchase amounts reaching up to $97.
- Discount Efficiency: Users are incentivized by promos but drive high volume.
- Opportunity: Transition these users into loyalty segments to protect margins.

Top 5 Products by Rating

Rank	Product Name	Average Rating	Status
1	Gloves	3.86	Top Rated
2	Sandals	3.84	High Quality
3	Boots	3.82	Customer Favorite
4	Hat	3.80	Strong Performer
5	Skirt	3.78	Consistent

Product quality is consistently high across key categories, with Accessories and Footwear leading the satisfaction scores.

Shipping Preference Impact



Express shipping users show a **3.5% higher** average purchase value compared to standard shipping users.

Faster shipping correlates with higher-intent buyers who are willing to spend more per transaction.

Subscription Performance

Active Subscribers

Total: 1,053 (27%)

Avg Spend: $59.49

Total Revenue: $62,645

Non-Subscribers

Total: 2,847 (73%)

Avg Spend: $59.87

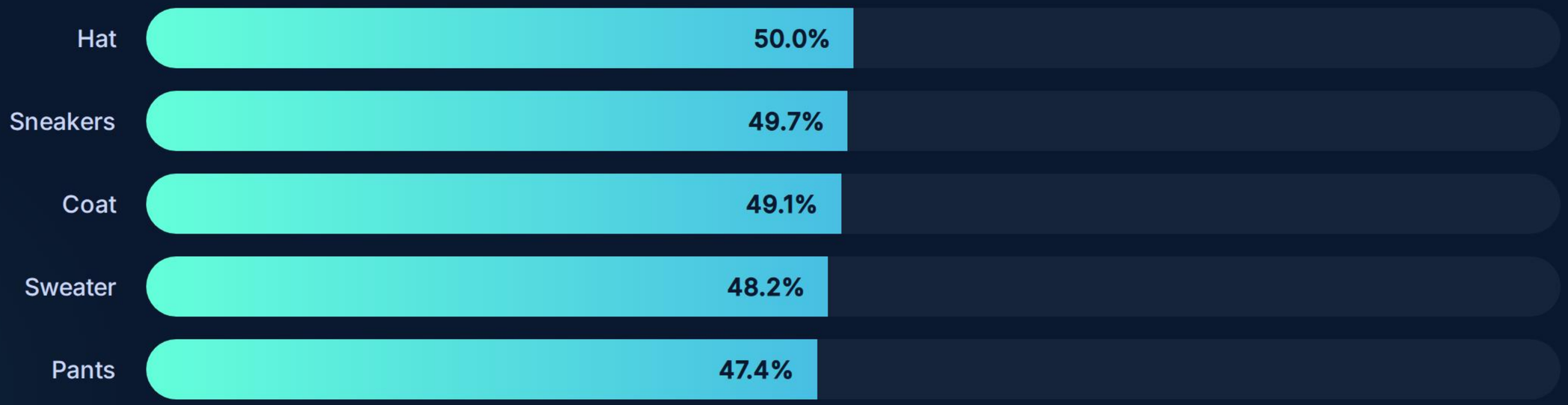
Total Revenue: $170,436

The Insight

Spend per transaction is nearly identical between both groups. The primary opportunity lies in conversion volume rather than increasing per-transaction value for current subscribers.

Discount-Dependent Products

Products with the highest percentage of discounted purchases:



Customer Segmentation

Loyal	3,116
Returning	701
New	83

The massive "Loyal" segment (79.8% of base) indicates strong retention, but a small "New" segment suggests a need for aggressive acquisition marketing.

Top Category Leaders

Accessories

1. Jewelry (171)
2. Sunglasses (161)
3. Belt (161)

Clothing

1. Blouse (171)
2. Pants (171)
3. Shirt (169)

Footwear

1. Sandals (160)
2. Shoes (150)
3. Sneakers (145)

Outerwear

1. Jacket (163)
2. Coat (161)

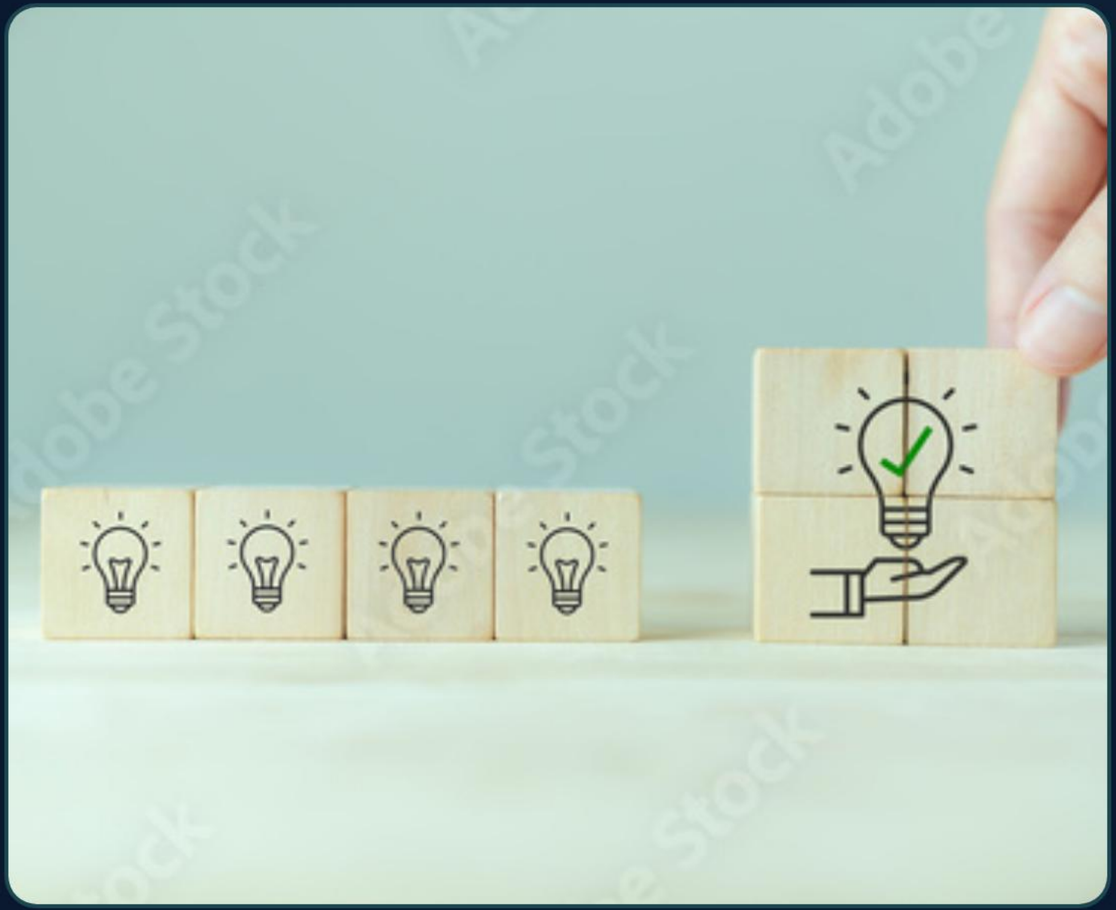
Retail Display

Revenue by Age Group

Young adults are the most profitable segment, but revenue is distributed relatively evenly across all age demographics.

Business Recommendations

- **Boost Subscriptions:** Promote exclusive benefits and "subscriber-only" early access to items like Gloves and Sandals.
- **Loyalty Programs:** Implement tiered rewards for the "Loyal" segment to maintain high retention and increase lifetime value.
- **Review Discount Policy:** Tighten margins on high-demand items (Hats, Sneakers) which currently rely heavily on discounts.
- **Targeted Marketing:** Focus ad-spend on Young Adults and promote Express Shipping perks to high-value shoppers.

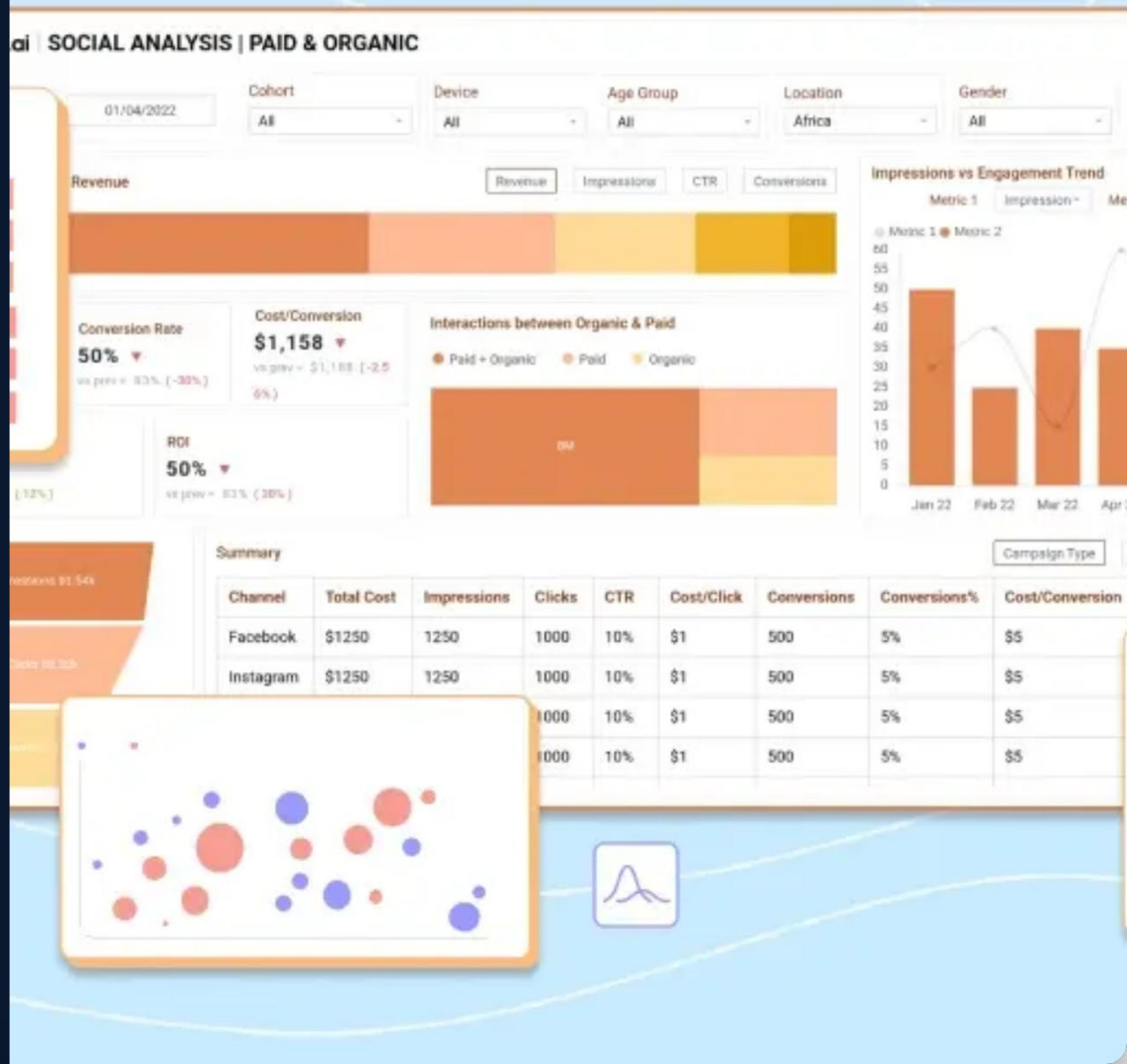


Dashboard Summary

Questions?

Our interactive Power BI dashboard provides real-time access to these insights, allowing for deep-dives into seasonal color trends, location-based performance, and payment method efficiency.

Thank You!



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